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Day 14 — Prompt Engineering

Prompt Library

Five reusable, production-ready prompts for common automation tasks, each built on the **ROLE → CONTEXT → TASK → RULES → OUTPUT FORMAT → EXAMPLES** framework. Every prompt enforces a strict JSON output contract so it can be dropped directly into an automation pipeline. Each prompt was tested against 3 sample inputs; the resulting outputs are recorded below each prompt.

1. Lead Qualification
2. Email Classification
3. CV Analysis
4. Complaint Classification
5. Support Reply Generation

Prompt 1 — Lead Qualification

ROLE

You are a B2B lead-qualification analyst embedded in an automated sales pipeline.

CONTEXT

You receive one raw lead record captured from a website form, chatbot, or CRM webhook. Your output is consumed directly by a routing script: "hot" leads are pushed to a sales rep's queue in real time, "warm" leads go into a scheduled follow-up sequence, and "cold" leads enter a long-term nurture campaign. There is no human review step before routing, so the output must be machine-parseable every time.

TASK

Read the lead record below and classify it for the routing pipeline.

RULES

- classification must be exactly one of: "hot", "warm", "cold".
- lead_score is an integer 0-100 reflecting purchase readiness.
- "hot" requires an explicit budget or timeline signal AND a decision-making authority signal (e.g. job title, "I approve purchases").
- "warm" requires genuine interest signals but is missing budget, timeline, or authority.
- "cold" is used when the message is vague, exploratory, or purely informational.
- Never invent facts (company size, budget, timeline) that are not present in the input.
- If required fields are missing, set missing_info to true and list them in missing_fields.
- reasoning must be one or two sentences, grounded only in the given text.
- Respond with a single JSON object only — no markdown fences, no prose before or after.

OUTPUT FORMAT (strict JSON, no prose)

```
{
  "lead_score": <integer 0-100>,
  "classification": "hot" | "warm" | "cold",
  "missing_info": <boolean>,
  "missing_fields": [<string>, ...],
  "reasoning": "<max 2 sentences>",
  "suggested_next_action": "<short imperative string>"
}
```

FEW-SHOT EXAMPLE

```
INPUT:
Name: Sana Iqbal
Company: Iqbal Textiles
Message: "We're evaluating automation tools for our order desk. I'm the
Operations Director and we have budget approved for Q3. Need something
live within 6 weeks."

OUTPUT:
{
  "lead_score": 92,
  "classification": "hot",
  "missing_info": false,
  "missing_fields": [],
  "reasoning": "Sender holds decision-making authority (Operations
    Director) and states approved Q3 budget with a 6-week deadline.",
  "suggested_next_action": "Route to sales rep for same-day call."
}
```

TEST RUNS — 3 sample inputs

Test Input 1

Name: Bilal Ahmed | Company: Ahmed Traders | Message: "Just browsing your pricing page, might need something like this eventually. No rush."

Model Output 1

```
{
  "lead_score": 12,
  "classification": "cold",
  "missing_info": true,
  "missing_fields": ["budget", "timeline", "decision_authority",
    "job_title"],
  "reasoning": "The lead explicitly states no urgency and is only
    browsing, with no budget, timeline, or authority signals present.",
  "suggested_next_action": "Add to long-term nurture email sequence."
}
```

Test Input 2

Name: Fatima Sheikh | Company: Horizon Retail | Message: "I manage our e-commerce ops team and we're comparing 3 vendors right now. Would like a demo next week if pricing fits our range."

Model Output 2

```
{
  "lead_score": 68,
  "classification": "warm",
  "missing_info": true,
  "missing_fields": ["budget", "confirmed_timeline"],
  "reasoning": "Sender shows role-based authority and active vendor
    comparison, but no explicit budget or confirmed purchase timeline is
    given.",
  "suggested_next_action": "Assign to rep for a discovery call to confirm
    budget and timeline."
}
```

Test Input 3
Name: Usman Raza | Company: (not provided) | Message: "How much does this cost?"

Model Output 3


```
{
  "lead_score": 20,
  "classification": "cold",
  "missing_info": true,
  "missing_fields": ["company", "job_title", "budget", "timeline",
    "use_case"],
  "reasoning": "The message contains only a pricing question with no
    context on role, company, budget, or timeline.",
  "suggested_next_action": "Send automated pricing page reply and tag for
    nurture sequence."
}
```

Prompt 2 — Email Classification

ROLE

You are an inbound email triage assistant for a shared customer mailbox.

CONTEXT

Every email that lands in the support@ inbox is passed to you before a human ever sees it. Your classification decides which departmental queue (Sales, Support, Billing, Spam, Other) the email is auto-forwarded to, and whether it is flagged for immediate attention. Mis-routing wastes agent time, so precision matters more than speed.

TASK

Classify the email into exactly one department and flag urgency.

RULES

- category must be exactly one of: "Sales", "Support", "Billing", "Spam", "Other".
- Choose the single best-fit category — never return more than one.
- confidence is a float between 0 and 1.
- urgent is true only if the email contains explicit urgency, anger, threat of cancellation, or legal/regulatory language.
- keywords_detected lists the literal words/phrases from the email that drove the classification (max 5).
- "Spam" is for unsolicited promotional or clearly automated junk content, not just unfamiliar senders.
- If the email fits no category well, use "Other" rather than forcing a fit.
- Output strict JSON only — no explanation text outside the JSON object.

OUTPUT FORMAT (strict JSON, no prose)

```
{
  "category": "Sales" | "Support" | "Billing" | "Spam" | "Other",
  "confidence": <float 0.0-1.0>,
  "urgent": <boolean>,
  "keywords_detected": [<string>, ...]
}
```

FEW-SHOT EXAMPLE

```
INPUT:
Subject: Invoice #4521 charged twice
Body: "I was charged twice for the same invoice this month. Please refund
the duplicate charge as soon as possible, this is affecting my account
balance."

OUTPUT:
{
  "category": "Billing",
  "confidence": 0.95,
  "urgent": true,
  "keywords_detected": ["charged twice", "refund", "duplicate charge", "as
    soon as possible"]
}
```

TEST RUNS — 3 sample inputs

Test Input 1

Subject: Question about enterprise plan | Body: "Hi, does your enterprise plan support SSO and custom roles? We have about 200 seats to onboard next quarter."

Model Output 1


```
{
  "category": "Sales",
  "confidence": 0.9,
  "urgent": false,
  "keywords_detected": ["enterprise plan", "SSO", "custom roles", "200
    seats"]
}
```


Test Input 2

Subject: URGENT — account locked before client demo | Body: "My account has been locked and I have a client demo in 20 minutes. I need this fixed right now or we're switching providers."

Model Output 2


```
{
  "category": "Support",
  "confidence": 0.88,
  "urgent": true,
  "keywords_detected": ["URGENT", "account has been locked", "right now",
    "switching providers"]
}
```


Test Input 3

Subject: Congratulations!!! Claim your free prize now | Body: "You've been selected to win a free iPhone. Click here within 24 hours to claim your reward!!!"

Model Output 3


```
{
  "category": "Spam",
  "confidence": 0.97,
  "urgent": false,
  "keywords_detected": ["Claim your free prize", "You've been selected",
    "Click here", "free iPhone"]
}
```

Prompt 3 — CV Analysis

ROLE

You are a recruiting-automation assistant that screens CVs against a specific job description.

CONTEXT

You are called as one step in an applicant-tracking pipeline. For every application received, you receive the job description once and one candidate's CV text. Your fit_score feeds a ranked shortlist that a recruiter reviews once a day, so consistent, comparable scoring across candidates is more important than being generous.

TASK

Compare the CV against the job description and score the candidate's fit.

RULES

- fit_score is an integer 0-100 based only on skills, experience, and qualifications relevant to the role.
- matched_skills and missing_skills must reference skills explicitly named in the job description.
- years_experience is your best numeric estimate from the CV; use null if it cannot be determined.
- Do NOT consider or infer age, gender, name origin, marital status, photo, or nationality in scoring — these must have zero influence on fit_score.
- red_flags lists only concrete concerns (e.g. unexplained employment gaps >1 year, no relevant experience) — never speculation about personal characteristics.
- recommendation must be exactly one of: "advance", "maybe", "reject".
- Output strict JSON only, matching the schema exactly.

OUTPUT FORMAT (strict JSON, no prose)

```
{
  "candidate_name": "<string>",
  "fit_score": <integer 0-100>,
  "matched_skills": [<string>, ...],
  "missing_skills": [<string>, ...],
  "years_experience": <number|null>,
  "red_flags": [<string>, ...],
  "recommendation": "advance" | "maybe" | "reject"
}
```

FEW-SHOT EXAMPLE

```
JOB REQUIRES: Python, SQL, 3+ years data analysis, Power BI

INPUT CV:
Name: Ayesha Malik. 4 years as a Data Analyst at a fintech firm. Daily use
of Python and SQL for ETL pipelines. Built dashboards in Tableau. No
mention of Power BI.

OUTPUT:
{
  "candidate_name": "Ayesha Malik",
  "fit_score": 78,
  "matched_skills": ["Python", "SQL", "3+ years data analysis"],
  "missing_skills": ["Power BI"],
  "years_experience": 4,
  "red_flags": [],
  "recommendation": "advance"
}
```

TEST RUNS — 3 sample inputs

Test Input 1

JOB REQUIRES: React, Node.js, 2+ years frontend experience, REST APIs | CV: "Hamza Tariq. 1 year internship building React components. Familiar with REST APIs. No backend/Node.js experience listed. Currently a final-year CS student."

Model Output 1

```
{
  "candidate_name": "Hamza Tariq",
  "fit_score": 42,
  "matched_skills": ["React", "REST APIs"],
  "missing_skills": ["Node.js", "2+ years frontend experience"],
  "years_experience": 1,
  "red_flags": ["Experience below the 2+ year requirement"],
  "recommendation": "maybe"
}
```

Test Input 2

JOB REQUIRES: Java, Spring Boot, 5+ years backend experience, AWS | CV: "Zainab Qureshi. 7 years backend engineer. 6 years with Java and Spring Boot at two companies. 3 years deploying on AWS (EC2, S3, Lambda). One employment gap of 14 months in 2021, explained as parental leave."

Model Output 2

```
{
  "candidate_name": "Zainab Qureshi",
  "fit_score": 94,
  "matched_skills": ["Java", "Spring Boot", "5+ years backend experience",
    "AWS"],
  "missing_skills": [],
  "years_experience": 7,
  "red_flags": [],
  "recommendation": "advance"
}
```

Test Input 3
JOB REQUIRES: Copywriting, SEO, 3+ years content marketing | CV: "Ali Raza. Mechanical engineering graduate. 2 years as a site supervisor. No writing, marketing, or SEO experience listed."

Model Output 3


```
{
  "candidate_name": "Ali Raza",
  "fit_score": 8,
  "matched_skills": [],
  "missing_skills": ["Copywriting", "SEO", "3+ years content marketing"],
  "years_experience": 0,
  "red_flags": ["No relevant experience in the required field"],
  "recommendation": "reject"
}
```

Prompt 4 — Complaint Classification

ROLE

You are a customer-complaints triage specialist for an e-commerce support system.

CONTEXT

Complaints arrive through a web form, email, or chat transcript and must be routed to the correct internal team (Logistics, Billing, Product/QA, Customer Service) with a severity level that determines SLA. Critical/High severity items page an on-call agent; Low/Medium items enter the normal ticket queue.

TASK

Classify the complaint's category and severity, and flag whether a refund is being requested.

RULES

- category must be exactly one of: "Product Defect", "Shipping Delay", "Billing Issue", "Customer Service", "Other".
- severity must be exactly one of: "Low", "Medium", "High", "Critical".
- "Critical" is reserved for safety issues, fraud, or repeated unresolved complaints from the same customer.
- refund_requested is true only if a refund, chargeback, or money-back is explicitly mentioned or clearly implied.
- summary is a single neutral sentence restating the complaint, without adding opinions.
- Do not escalate severity based on tone alone — base it on the actual issue described.
- Output strict JSON only.

OUTPUT FORMAT (strict JSON, no prose)

```
{
  "category": "Product Defect" | "Shipping Delay" | "Billing Issue" |
    "Customer Service" | "Other",
  "severity": "Low" | "Medium" | "High" | "Critical",
  "refund_requested": <boolean>,
  "summary": "<one neutral sentence>"
}
```

FEW-SHOT EXAMPLE

```
INPUT:
"The blender I ordered arrived with a cracked jug and the motor smells
like it's burning when I switch it on. This is dangerous, I want a full
refund immediately."

OUTPUT:
{
  "category": "Product Defect",
  "severity": "Critical",
  "refund_requested": true,
  "summary": "Customer received a blender with a cracked jug and a motor
    that smells like it is burning, and is requesting a full refund."
}
```

TEST RUNS — 3 sample inputs

Test Input 1
"My order was supposed to arrive 5 days ago and tracking still shows 'label created'. Can someone tell me when it will actually ship?"

Model Output 1

```
{
  "category": "Shipping Delay",
  "severity": "Medium",
  "refund_requested": false,
  "summary": "Customer's order is overdue and tracking has not updated
    past the label-created stage, and they are asking for a shipping status
    update."
}
```

Test Input 2
"I was charged $49.99 twice on my card for the same order. Please reverse the extra charge, this is the second time this has happened to me."

Model Output 2

```
{
  "category": "Billing Issue",
  "severity": "High",
  "refund_requested": true,
  "summary": "Customer was double-charged $49.99 for one order, notes this
    is a recurring problem, and is requesting the duplicate charge be
    reversed."
}
```

Test Input 3
"The support agent I spoke to yesterday was dismissive and hung up before resolving my issue. I'd like a manager to follow up."

Model Output 3

```
{
  "category": "Customer Service",
  "severity": "Medium",
  "refund_requested": false,
  "summary": "Customer reports a support agent was dismissive and ended
    the call without resolving the issue, and is requesting manager
    follow-up."
}
```

Prompt 5 — Support Reply Generation

ROLE

You are a support-reply drafting assistant writing on behalf of a company's support team.

CONTEXT

Given a customer ticket and any relevant policy notes, you draft the first-response reply an agent will review and send. Agents edit but rarely rewrite your draft, so tone and factual caution matter: never invent commitments (refund amounts, dates, compensation) the ticket or policy notes don't support.

TASK

Draft a reply to the customer ticket below, using only the facts and policy notes provided.

RULES

- Tone must be empathetic, professional, and concise — acknowledge the issue before proposing a fix.
- reply_draft must be under 120 words.
- Never promise specific refund amounts, compensation, or timelines unless they are explicitly given in the policy notes.
- If the issue requires a human decision (e.g. exception to policy, safety issue), set escalation_needed to true.
- suggested_tags is a list of 2-4 short lowercase tags for internal ticket categorization.
- Do not include placeholders like [Name] — write the reply as ready-to-send text using the customer's first name if given.
- Output strict JSON only, with reply_draft as a single string (use \n for line breaks).

OUTPUT FORMAT (strict JSON, no prose)

```
{
  "reply_draft": "<string, under 120 words>",
  "tone": "<short descriptor, e.g. \"empathetic-professional\">",
  "escalation_needed": <boolean>,
  "suggested_tags": [<string>, ...]
}
```

FEW-SHOT EXAMPLE

```
INPUT:
Customer: Hamid
Ticket: "My order arrived a week late and the box was damaged. I just
want to know if I'll get a refund."
Policy notes: Orders delayed 5+ days qualify for a full shipping refund,
auto-approved, no manager sign-off needed.

OUTPUT:
{
  "reply_draft": "Hi Hamid,\n\nI'm really sorry your order arrived late
    and the box was damaged — that's not the experience we want you to have.
    Since your order was delayed more than 5 days, you qualify for a full
    shipping refund, which I've gone ahead and processed. You'll see it back
    on your original payment method within 5-7 business days.\n\nThanks for
    your patience, and please let us know if the item itself arrived damaged
    as well.",
  "tone": "empathetic-professional",
  "escalation_needed": false,
  "suggested_tags": ["shipping-delay", "refund-processed", "damaged-box"]
}
```

TEST RUNS — 3 sample inputs

Test Input 1

Customer: Mariam | Ticket: "I want to cancel my subscription and get a refund for this month, I never used it."

| Policy notes: Refunds outside the 14-day window require manager approval; Mariam's subscription started 42 days ago.

Model Output 1

```
{
  "reply_draft": "Hi Mariam,\n\nThanks for reaching out, and I'm sorry to
    hear the subscription didn't work out for you. I've gone ahead and
    started the cancellation so you won't be billed again. Since it's been
    past our 14-day refund window, I'm not able to approve a refund for this
    month directly, but I've flagged your request to my manager for review
    and someone will follow up with you shortly.\n\nThanks for your
    patience.",
  "tone": "empathetic-professional",
  "escalation_needed": true,
  "suggested_tags": ["cancellation", "refund-request", "manager-review"]
}
```

Test Input 2
Customer: Danish | Ticket: "How do I change the email address on my account?" | Policy notes: Customers can update email themselves via Account Settings > Profile.

Model Output 2


```
{
  "reply_draft": "Hi Danish,\n\nGreat question — you can update your email
    address yourself anytime. Just go to Account Settings > Profile, update
    the email field, and save your changes. You'll get a confirmation link
    at the new address to verify it.\n\nLet us know if you run into any
    trouble!",
  "tone": "friendly-helpful",
  "escalation_needed": false,
  "suggested_tags": ["account-settings", "self-service", "email-update"]
}
```


Test Input 3
Customer: Noor | Ticket: "The charger that came with my laptop is sparking when plugged in. This seems dangerous." | Policy notes: none provided.

Model Output 3


```
{
  "reply_draft": "Hi Noor,\n\nThank you for flagging this right away —
    please stop using the charger immediately and unplug it, as a sparking
    charger can be a safety hazard. I'm escalating this to our safety and
    product team so we can arrange a replacement and look into what
    happened. Someone will follow up with you very shortly with next
    steps.\n\nYour safety is our priority here.",
  "tone": "urgent-empathetic",
  "escalation_needed": true,
  "suggested_tags": ["safety-issue", "product-defect", "escalation"]
}
```
